

① a) $\lim_{n \rightarrow \infty} \frac{n}{2n+1} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2} \therefore \text{converge}$ b) $\lim_{n \rightarrow \infty} 8+2n = \lim_{n \rightarrow \infty} 8 + \lim_{n \rightarrow \infty} 2n = \infty \therefore \text{Diverge}$

② $\sum_{n=1}^{\infty} \frac{n^2}{(n!)^{n+1}}$

③ * converge absolutamente se $\sum |a_n|$ converge

Testar $|a_n|$

* Teste do n-ésimo termo: $\lim_{n \rightarrow \infty} \frac{n}{n^2+1} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0$ Conclusão.

* Teste do raio: $\lim_{n \rightarrow \infty} \frac{\frac{n+1}{(n+1)^2+4}}{\frac{n}{n^2+4}} = \lim_{n \rightarrow \infty} \frac{n+1}{(n+1)+4} \cdot \lim_{n \rightarrow \infty} \frac{n^3+4n^2+4}{n^3+2n^2+5n} = \lim_{n \rightarrow \infty} \frac{1+\frac{1}{n}+\frac{4}{n^2}}{1+\frac{2}{n}+\frac{5}{n^2}} = 1$ Conclusão.

* Teste de integral: $\int_1^{\infty} \frac{x}{x^2+4} dx = \int_1^{\infty} \frac{\frac{1}{u}}{u} \frac{du}{2} = \left[\ln u \right]_5^{\infty} = \left[\ln x^2+4 \right]_5^{\infty} = \infty \therefore \text{diverge}$

$u = x^2+4 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2}$

Testar a_n

$\lim_{n \rightarrow \infty} \frac{(-1)^{n-1} \cdot n}{n^2+4} = \frac{n \cdot (-1)^{n-1}}{n(n+4)} = 0$

$\frac{(n+1)}{(n+1)^2+4} \leq \frac{n}{n^2+4} \rightarrow 4 \leq n^2+n$

$\frac{+}{-2-\sqrt{n+1}} \quad \frac{-}{-2+\sqrt{n+1}}$

Valido $\forall n \geq 2$

* converge

$\therefore$ A série é condicionalmente convergente

④ $\frac{3^1}{10^1} + \frac{3^2}{10^2} + \frac{3^3}{10^3} \dots \rightarrow \frac{1}{10} = u \quad S_n = \frac{3^1}{10^1} = \frac{3^1}{100} = \frac{3^1}{99} + 5 = \frac{532}{99}$

$S_{\infty} = \frac{3^1}{99} + 5 = \frac{532}{99}$

⑤ a) $\frac{1}{(n+2)(n+3)} = \frac{A}{(n+2)} + \frac{B}{(n+3)} = \frac{A(n+3)+B(n+2)}{(n+2)(n+3)} \quad A(n+3)+B(n+2)=1 \quad n=-3 \quad B=-1 \quad n=-2 \quad A=1$

$\sum_{n=1}^{\infty} \frac{1}{(n+2)(n+3)} = \left\{ \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) \dots \right\} \quad S_{\infty} = \frac{1}{3}$

b) $\sum_{n=5}^{\infty} \left(\frac{e}{11} \right)^{n-1}$ * Série geométrica:

$\frac{e}{11} < 1 \therefore \text{converge}$

$S = \frac{a_1}{1-r} \rightarrow S = \frac{\left(\frac{e}{11} \right)^4}{1 - \frac{e}{11}} = \frac{e^4}{11^4(11-e)}$

⑥ a) $\sum_{n=1}^{\infty} \frac{4^{n+2}}{7^{n-1}}$ * Teste do raio: $\lim_{n \rightarrow \infty} \frac{4^{n+3}}{4^{n+2}} = \lim_{n \rightarrow \infty} \frac{4}{7} = \frac{4}{7} < 1 \therefore \text{converge}$

b) $\sum_{n=2}^{\infty} \frac{n}{n^2-5} \rightarrow$ * Teste do n-ésimo termo $\lim_{n \rightarrow \infty} \frac{n}{n^2-5} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{2n-5} = 0$ Conclusão

* Teste de comparação de limites:

$b_n = \frac{1}{n} \rightarrow \text{converge} \quad \lim_{n \rightarrow \infty} \frac{n}{n^2-5} \cdot \frac{n^2}{n^2} = \lim_{n \rightarrow \infty} \frac{n^3}{n^2(1-\frac{5}{n^2})} = 1 \therefore \text{converge}$

d) $\sum_{n=1}^{\infty} \frac{n}{1+n^2}$

* n-ésimo termo

$\lim_{n \rightarrow \infty} \frac{n}{1+n^2} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{2n} = \frac{1}{2} \therefore \text{diverge}$

c) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot n!}{(2n-1)!}$ * Teste do raio: $\lim_{n \rightarrow \infty} \frac{(n+1)!}{(2n+1)(2n)!} = \lim_{n \rightarrow \infty} \frac{n+1}{2n^2+2n} \stackrel{L'H}{=} \lim_{n \rightarrow \infty} \frac{1}{4n+2} = 0 < 1 \therefore \text{converge}$

